



CGM Data and Report Interpretation



Device download reports vary by manufacturer and include differing data presentations.

Written by: danatech staff and reviewed by clinical members of the ADCES Diabetes Tech Committee.

Part of danatech from ADCES's commitment to improving equitable access and appropriate use of diabetes tech in clinical practice.

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Worksheets

- [WORKSHEET 7: ICC Framework](#)
- [Worksheet 11: Sample Instructions for Professional CGM for the Person with Diabetes](#)
- [WORKSHEET 11: Sample Instructions for Personal CGM for the Person with Diabetes](#)
- [WORKSHEET 12: Basic and Advanced Personal CGM Training Needs \(Personal CGM\)](#)

CGM Report Evaluation

The health care professional interprets the personal CGM reports retrospectively and evaluates for glycemic excursions above/below target range, seeking to identify patterns and potential causes for these excursions with the person with diabetes.

Device download reports vary by manufacturer and include differing data presentations; however, there has been a recent push to focus on use of the Ambulatory Glucose Profile (AGP) report, which was developed by the Park Nicollet International Diabetes Center (IDC) in Minneapolis, Minnesota²⁷. The use of one report would aid in standardization of care and would help to make interpretation more accurate and efficient. The AGP report includes summary statistics, a glucose profile graph and an insulin profile graph or glucose daily calendar graphs.

EXAMPLE OF AGP REPORT

AGP Report

GLUCOSE STATISTICS AND TARGETS

26 Feb 2019 - 10 Mar 2019

% Time CGM is Active

13 days

99.9%

Glucose Ranges

Targets [% of Readings (Time/Day)]

Target Range 70-180 mg/dL

Greater than 70% (16hr 48min)

Below 70 mg/dL

Less than 4% (58min)

Below 54 mg/dL

Less than 1% (14min)

Above 250 mg/dL

Less than 5% (1hr 12min)

Each 5% increase in time in range (70-180 mg/dL) is clinically beneficial.

Average Glucose

173 mg/dL

Glucose Management Indicator (GMI)

7.6%

Glucose Variability

49.5%

Defined as percent coefficient of variation (%CV); target ≤36%

Name

MRN

TIME IN RANGES

Very High (>250 mg/dL)

20% (4hr 48min)

High

23% (5hr 31min)

Target Range (70-180 mg/dL)

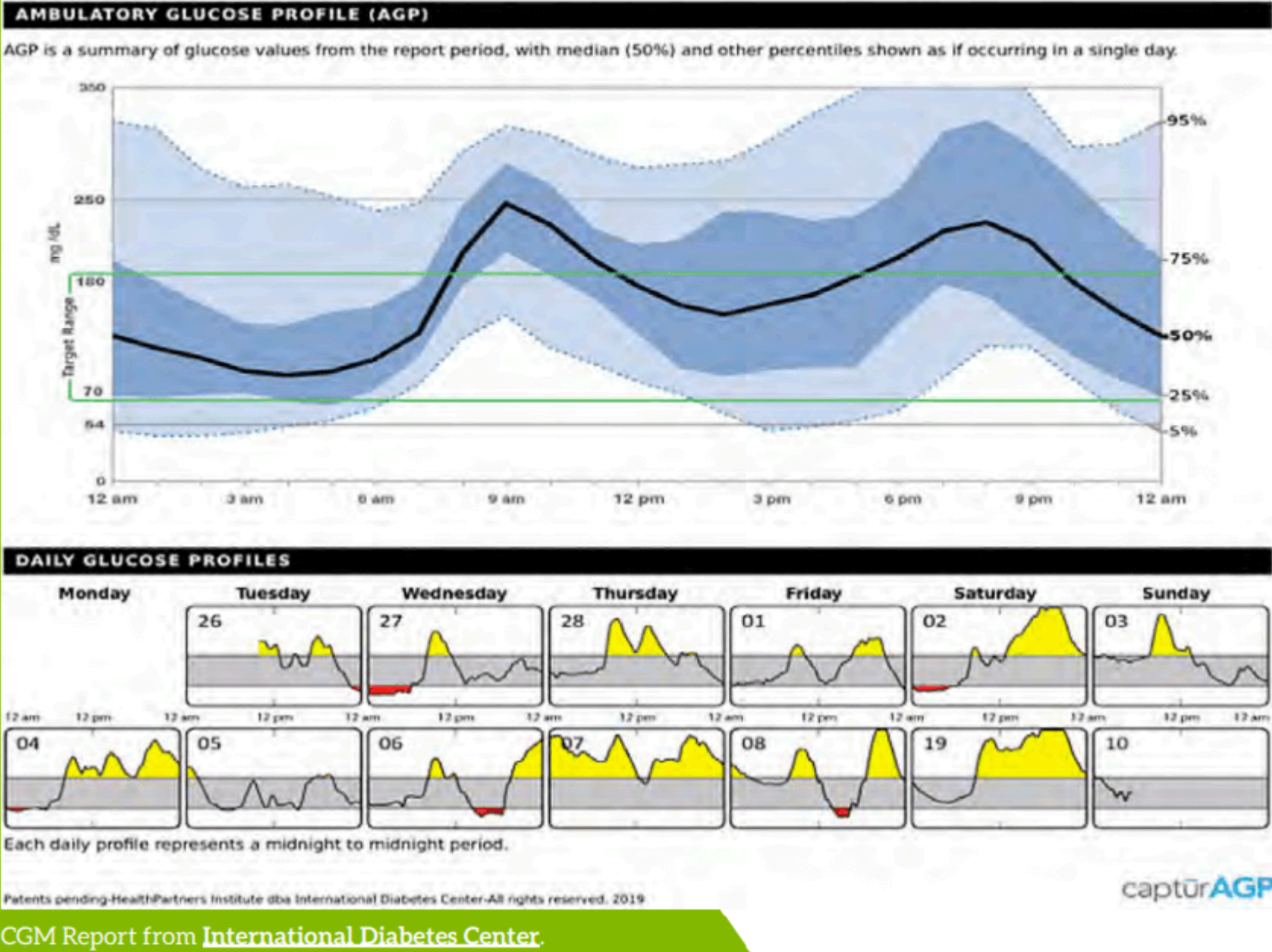
47% (11hr 17min)

Low

4% (58min)

Very Low (<54 mg/dL)

6% (1hr 26min)



Coefficient of Variation (CV) This is a measure of glycemic variability. A CV of less than or equal to 36% is considered acceptable. A CV greater than 36% is considered unstable and intervention is needed.

Very High Time Above Range (TAR) % of readings and time >250 mg/dl

High Time Above Range (TAR) % of readings and time 181-250 mg/dl

Time In Range (TIR) % of readings and time 70-180 mg/dl

Low Time Below Range (TBR) % of readings and time 54-69 mg/dl

Very Low Time Below Range (TBR) % of readings and time <54 mg/dl

Glucose Time In Range Measure

Glucose time in target range (TIR) is another important measure and Dr. Richard Bergenstal, in 2018, identified correlations of TIR and HbA1C as follows: (Used with permission from Sage Publications. Reference Guide for Integrating Continuous Glucose Monitoring Into Clinical Practice³¹)

TIME IN RANGE % (TIR%)	AVERAGE HBA1C
40%	8.1%
50%	7.7%
60%	7.3%
70%	6.9%
80%	6.5%

Step 1 Confirm that adequate data are available. For Current CGM users, a minimum of 70% of 2 weeks of data is recommended.³⁴ Fewer days are needed when professional CGM systems are used.

Step 2 Print out the AGP and ask patients to describe their daily self-management. When are they taking their insulin and how much? When do they wake? When do they eat? Do they exercise and, if so, what type of exercise and when are they doing it? Document this information on the AGP printout.

Step 3 Ask the patients what they see in the AGP and why they think it may be important. Then listen. Interactive discussion with patients allows them to better understand how their insulin, food and other factors affect their glucose levels and also helps clinicians identify knowledge deficits or behaviors that may not support glycemic goals.

Step 4 Look for problematic glycemic patterns in the following order of priority:

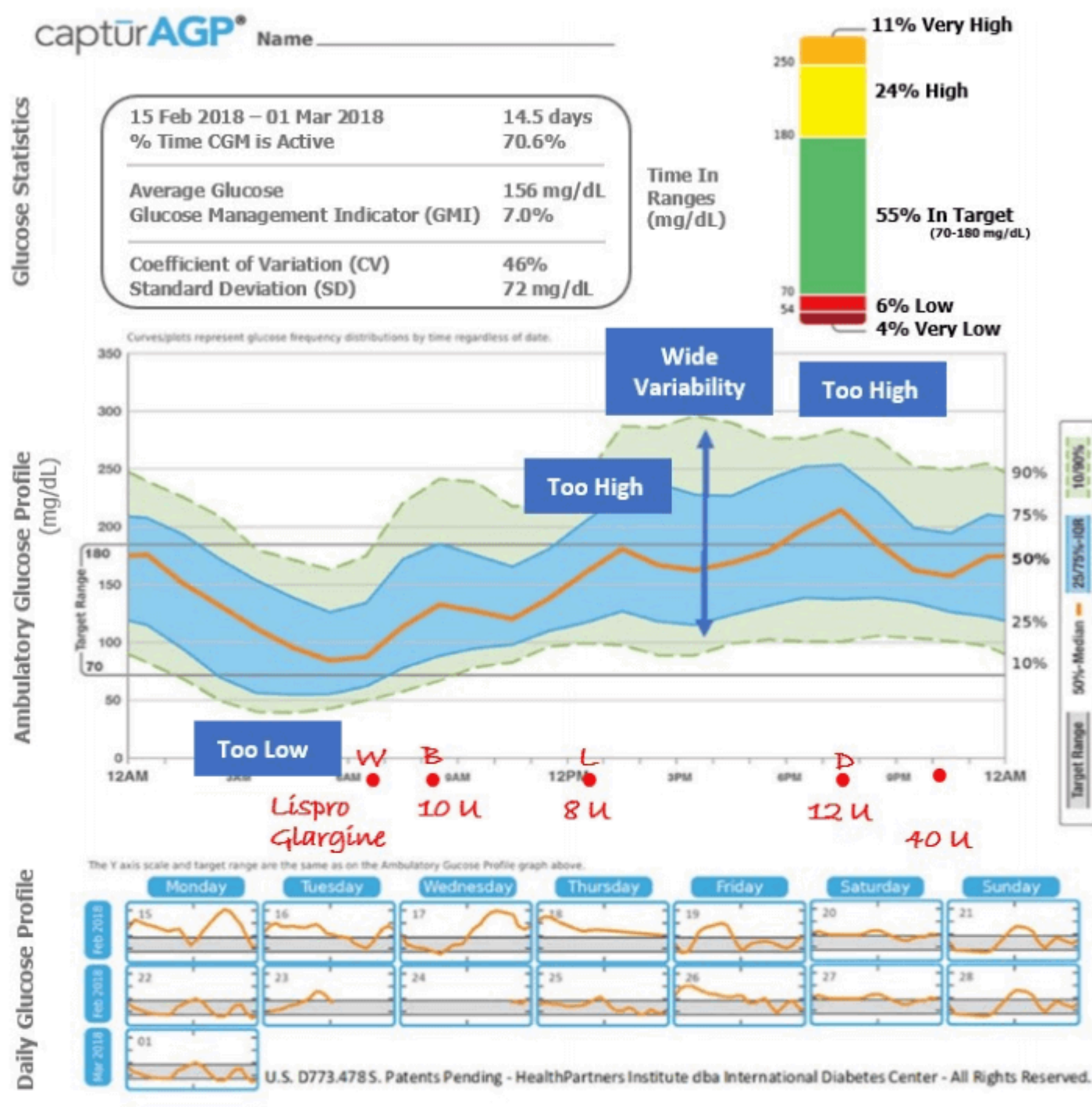
- 1. Hypoglycemia
- 2. Hyperglycemia
- 3. Wide glycemic variability.

Review the overall glucose profile (initial view) to determine the time of day when patterns are occurring, then review the daily graphs to double-check patterns to see if they are clustered on certain days.

Step 5 Encourage patients to reflect on what they think may be causing the problem and discuss potential solutions.

Step 6 Collaboratively develop an action plan. Make sure patients fully understand the changes they will be making and that they have the knowledge/skills to implement the plan.

Step 7 Make a copy of the marked up AGP printout for the patient and enter the original into the electronic medical record (EMR). If electronic entry is not possible, copy and paste the AGP into the EMR as a progress note.



History:

- Robert is 27-year-old male who has had T1DM for 8 years, treated with multiple daily insulin injection (MDI) therapy-aspart at meals.
- His self-reported insulin regimen is 8-12U aspart at meals, 40U glargine at bedtime.
- His current A1C (7.0%) is at target; however, Robert complains of occasional headaches when he wakes in the morning.

Interpretation:

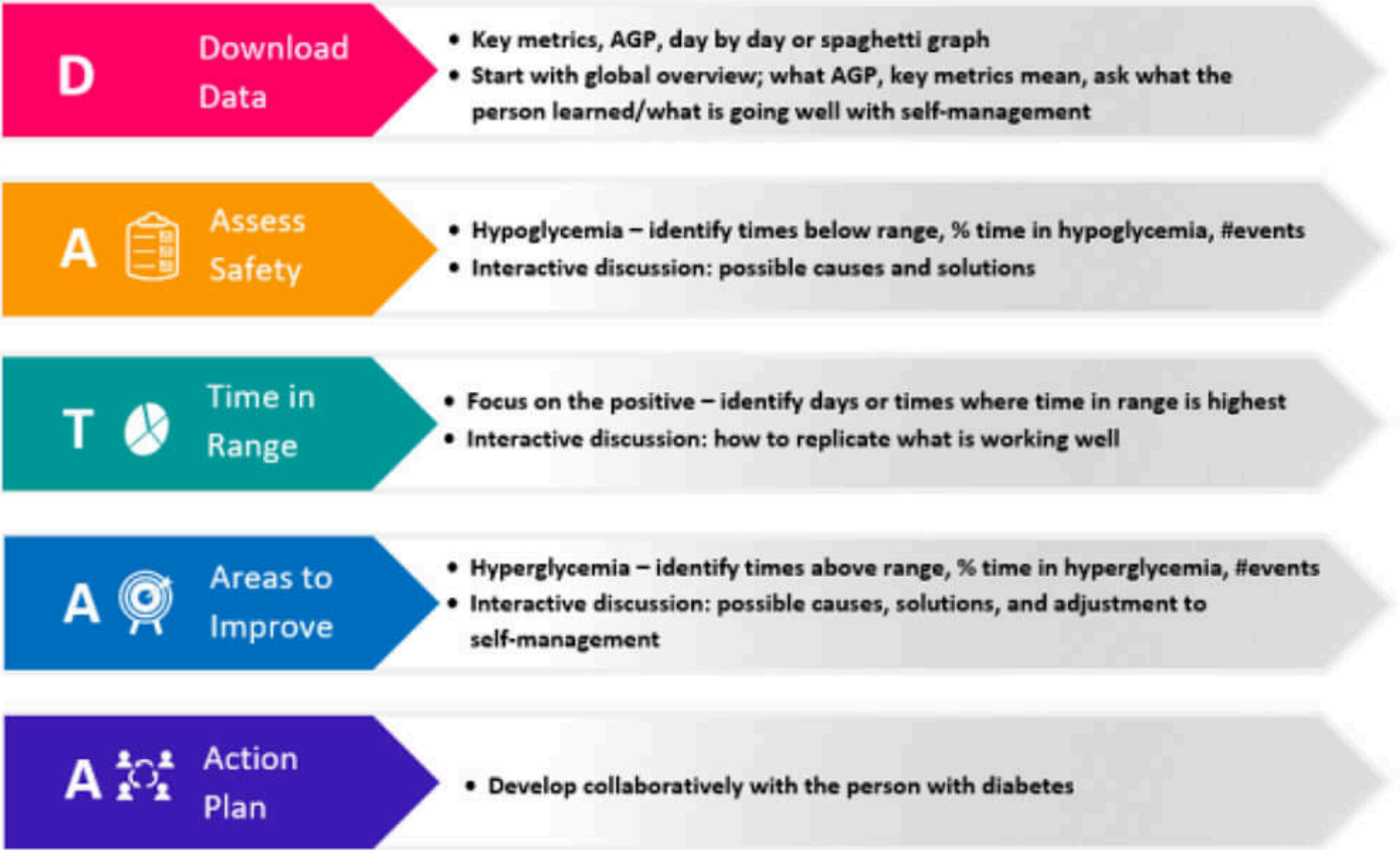
- AGP indicates an adequate number of days (14) for assessment.
- Extremely low glucose for several hours in the early morning.
- High glucose post-supper appears to be driven by post-lunch excursions.
- Glycemic variability indicates instability.
- Daily Graphs suggest that Sunday mornings are problematic for lows; however glycemic patterns vary from day to day.
- Wide glycemic variability prompts discussion about meals/snacks and insulin dosing. Robert explains that he does snack throughout the day but is uncomfortable about calculating insulin doses; he avoids taking correction insulin even when glucose is high.

Action:

- Reduce the glargine dose at bedtime to address the low glucose. Another option is to switch the patient to glargine U-300 or degludec. This would avoid possibility of elevated afternoon glucose.
- Provide/schedule education in CHO counting, insulin dose calculation and use of correction insulin.
- Review CGM data again at 2 weeks - if overnight glucose is stabilized, increase the aspart insulin at lunch and supper to address the postprandial glucose.

Additionally, ADCES recommends the use of the DATAA Model for reviewing CGM data, as described by Isaacs et al in the August 2020 issue of "The Diabetes Educator"³²:

Review of CGM - DATAA



At each step, express that this is information, not good or bad

CGM Topics & Resources

- [Find & Compare CGM Devices](#)
- [Panther CGM Tools](#)
- [CGM 101, Getting Started](#)
- [Monitoring in Professional Practice](#)
- [Candidate Selection & Training for CGMs](#)
- [Device Affordability Programs](#)
- [Diabetes Tech Billing Codes & Reimbursement](#)

Glucose Monitoring Training & Education

- Integrating Diabetes Technology into Primary Care Part 1: [Overview and Clinical Scenarios](#)
- Integrating Diabetes Technology into Primary Care Part 2: [Interactive Case Studies](#)
- [Understand the Limitations of CGM Metrics: Nothing is Perfect Mastering CGM: PANTHER Program's Guide to Data Interpretation and Clinical Assessment](#)

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